



ABSTRACT

SBIO 221LB Human Anatomy & Physiology I Lab is a face-to-face introductory laboratory course designed for students in nursing, allied health sciences, nutrition, and other health-related majors. The initial course emphasized foundational anatomy and physiology concepts through gross and microscopic anatomy, anatomical models, tissue examination, microscopy, and EMG/EEG activities. The redesigned course integrates Science Gateways, Jupyter, BioDigital Human, and responsible AI tools to enhance student engagement, visualization, data analysis, and scientific communication while maintaining core course outcomes.

INTRODUCTION

The initial course provides a strong foundation in hands-on anatomy and physiology through microscopy, anatomical models, laboratory measurements, and practical activities. Building on this foundation, the redesigned course integrates Science Gateways, Jupyter, and AI to visualize physiological data, analyze functional relationships, and communicate scientific findings. This approach expands student inquiry, engagement, and technological literacy while aligning with Southern University’s commitment to excellence, innovation, and equity in STEM education.

Course Goals

- Strengthen anatomy-physiology structure-function relationships and homeostasis.
- Integrate Science Gateway resources, computational thinking, and responsible AI into lab learning.
- Expand student access to computing tools, biomedical data, visualization, and interactive hands-on learning.
- Provide faculty/students with training, technical support, and reusable instructional resources.

Grade Computation

|                                        |     |
|----------------------------------------|-----|
| 3 Laboratory Practicums                | 40% |
| Written Lab/Content Assignments        | 20% |
| In-Classroom & Demonstrated Activities | 20% |
| Gateway/AI Activities (Jupyter & Data) | 10% |
| Low-Stakes Graded Practice             | 10% |

SCIENCE GATEWAY NEEDS / QUESTIONS

1. How to balance computing support for students with varied experience levels?
2. Which Science Gateway resources are most effective for anatomy & physiology lab?
3. How to ensure accessibility (devices, accounts, internet)?
4. Best practices for integrating AI responsibly and ethically?
5. How to evaluate student learning and engagement with computational tools?

Candidate Gateway Ecosystem

| PhysioNet                                             | CyVerse                                                           |
|-------------------------------------------------------|-------------------------------------------------------------------|
| Open physiological signals for EMG / EEG exploration. | Browser-based notebooks, shared data, and reproducible workflows. |
| Galaxy                                                | Cell Collective                                                   |
| Accessible bioinformatics for large datasets.         | Interactive models of biological systems.                         |

IMPACT ON STUDENTS

Students are exposed to real data sets, visualize complex anatomical structures, analyze physiological signals, and develop computational thinking. The modified syllabus helps students interpret physiological signals, connect lab observations with broader decision-making, and access Science Gateway tools that build toward biomedical computing.

TARGET COURSE DESCRIPTION

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Course Name       | Human Anatomy & Physiology I Lab                                                   |
| Course Number     | SBIO 221LB                                                                         |
| Department        | Biology / Health Sciences                                                          |
| Enrollment        | Introductory undergraduate lab                                                     |
| Prerequisites     | Per program requirements / concurrent lecture if applicable                        |
| Project Resources | Gateways, Jupyter, BioDigital Human, CyVerse, PhysioNet, Galaxy, AI literacy tools |

Course Description

Human Anatomy & Physiology I Lab examines gross and microscopic anatomy and physiological principles associated with homeostasis. The modified course preserves face-to-face laboratory work while utilizing online learning, 3D visualization, data interpretation, responsible AI-supported learning, and computational resources for enhanced inquiry.

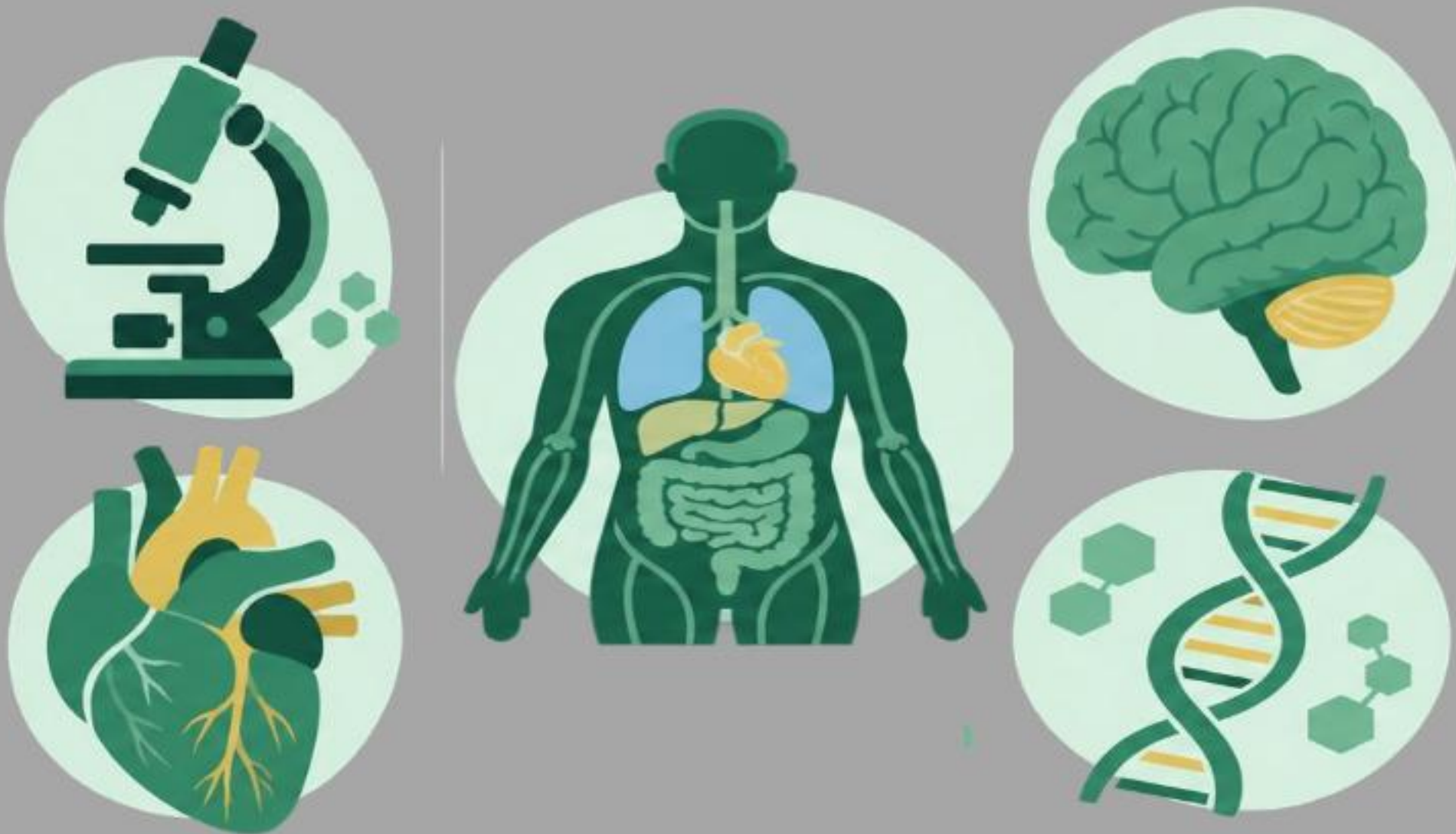
Key Content by Week

- Weeks 1-2: lab skills, orientation, anatomical language, microscopy, homeostasis, AI introduction.
- Weeks 4-7: tissues, organ systems, anatomical models, virtual anatomy platforms.
- Weeks 8-11: physiology lab activities, data collection (EMG/EEG), computational exploration.
- Weeks 12-15: integration, case studies, project work, data analysis, scientific communication.

SCIENCE GATEWAY RESOURCES USED

BioDigital Human -3D anatomy exploration and structure-function activities  
Jupyter Notebooks -interact with datasets, create visualizations, and interpret data  
CyVerse -cloud-based data storage, notebooks, analysis tools, and collaborative workflows  
PhysioNet -open physiological signal datasets for EMG/EEG exploration  
Galaxy / Cell Collective -optional browser-based tools for reproducible analysis and systems modeling

Gateway-Enhanced Learning Model



SIGNATURE GATEWAY ACTIVITIES / STUDENT DELIVERABLES

What students create, verify, visualize, analyze, and communicate.

Gateway-Enhanced Lab Portfolio

1. AI Anatomy Fact-Check: compare an AI explanation with course resources.
2. 3D Structure-Function Exploration: use BioDigital Human and physical models.
3. Clinical Case Challenge: apply lab concepts to a short health scenario.
4. Physiology Data Investigation: organize and visualize EMG or EEG data in a guided Jupyter notebook.
5. Create-to-Learn Product: submit an annotated diagram, concept map, or short scientific explanation.

Observe → Visualize → Analyze → Evaluate → Communicate

Represents the student learning process: observe anatomical and physiological evidence, visualize structures and data using digital tools, analyze patterns and relationships, evaluate findings for accuracy and meaning, and communicate evidence-based scientific conclusions.

Project Goals

- Demonstrate how Science Gateways support teaching and research activities in introductory health science courses.
- Expand student access to computing tools, data, and hands-on learning.
- Build a scalable model for faculty-driven training, technical support, and future course adaptations.

EXPANSION IDEAS

- Add additional Science Gateway tools.
- Incorporate authentic datasets and case-based learning.
- Explore interdisciplinary collaborations (nursing, public health, physiology).
- Develop open educational resources that can be shared across HBCUs.
- Support undergraduate research and course-based projects.

REFERENCES

1. BioDigital Human. (n.d.). <https://www.biodigital.com>
2. CyVerse. (n.d.). <https://cyverse.org>
3. PhysioNet. (n.d.). <https://physionet.org>
4. Galaxy Project. (n.d.). <https://usegalaxy.org>
5. iWorx Systems. (n.d.). <https://www.iworx.com>

LESSONS LEARNED

- Preserve hands-on microscopy, models, and practicum preparation.
- Add technology to improve 3D visualization and data literacy.
- Use low-cost tools, guided notebooks, and peer instruction to reduce barriers.
- Provide faculty development and institutional support for sustainable adoption.

COURSE REDESIGN: BEFORE & AFTER

A side-by-side look at the initial syllabus and what the redesign adds.

INITIAL SYLLABUS

- Face-to-face lab focused on microscopy, models, and core anatomy/physiology concepts.

- Limited use of computational tools.
- Traditional lab assignments and practicals.
- Assessment focused on lab practicums and written work.
- Limited opportunities for data visualization or AI literacy.

Course Integration Map

| Week  | Lab Focus                    | Gateway Integration     | Assessment    |
|-------|------------------------------|-------------------------|---------------|
| 1-3   | Lab skills & microscopy      | None (traditional)      | Practicum     |
| 4-7   | Tissues & organ systems      | BioDigital Human        | Lab report    |
| 8-11  | Physiology & data collection | Jupyter / iWorx EMG/EEG | Data analysis |
| 12-15 | Integration & projects       | Multiple gateways       | Presentation  |

Featured Gateway Exercise: EMG

Question: How does electrical activity reflect skeletal muscle function?

- 1 RECORD Perform the lab and collect EMG data.
- 2 ACCESS Open a Jupyter notebook and load the dataset.
- 3 VISUALIZE Plot electrical activity across time and conditions.
- 4 COMPARE Relate the local lab signal to reference data.
- 5 INTERPRET Connect motor-unitactivation to musclefunction.
- 6 COMMUNICATE Submit a figure with a short evidence-based explanation.

MODIFIED SYLLABUS

- Preserves the initial hands-on lab structure, major topics, and grading framework.
- Adds Science Gateway alignment to strengthen inquiry, visualization, analysis, evaluation, and communication.
- Integrates BioDigital Human, Jupyter, iWorx EMG/EEG, and AI literacy activities.
- Adds AI Anatomy Fact-Check, 3D Structure-Function Exploration, Clinical Case Challenge, Physiology Data Investigation, and Create-to-Learn Product.
- Expands access to computing tools, faculty development, and reusable Gateway-enhanced resources.

Skills to Scale

Adapt modules for A&P II and health sciences.

Expected Educational Impact

- Traditional Lab Skills
- Biomedical Data Literacy
- Computational Exploration
- Reproducible Science

Implementation Phases

- Phase 1 Pilot EMG + EEG gateway activities in SBIO 221LB.
- Phase 2 Refine Collect usability data and revise instructions.
- Phase 3 Expand Add tissue imaging and musculoskeletal modules.

AUTHORS & MENTORS

Gateway community & faculty support



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POSTER FOCUS

Higher education access, inclusive learning, STEM workforce development.



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